Exercise 2: Continuum Mechanics

Task 2.1: Mass Balance

An incompressible fluid flows through a constricting pipe as shown in Fig. 2.1. Derive a relationship between the flow velocities $\dot{\boldsymbol{x}}^{\mathrm{I}}$ and $\dot{\boldsymbol{x}}^{\mathrm{II}}$ at the inlet and outlet cross-sections, using the global form the mass balance in the current configuration:

$$\dot{m} = \int_{\mathcal{S}} \frac{\partial \rho}{\partial t} \, \mathrm{d}v + \int_{\partial \mathcal{S}} \rho \, \dot{\boldsymbol{x}} \cdot \boldsymbol{n} \, \mathrm{d}a = 0.$$

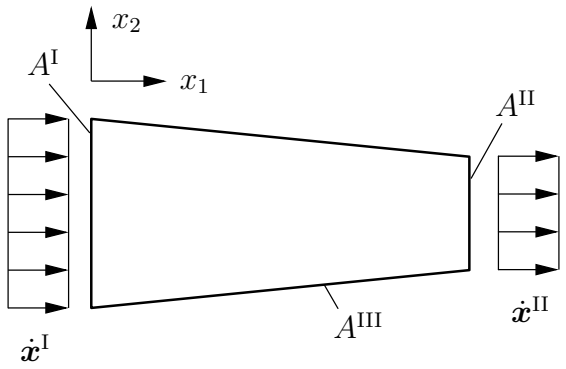

Figure 2.1: Schematic sketch of a narrowing pipe.

Task 2.2: Deformation Gradient and Strain Measure

Given below is the material description of a body's motion with the components

$$x_1 = X_1 + \alpha X_2, \qquad x_2 = X_2 + \beta X_3, \qquad x_3 = X_3 + \gamma X_1$$

in an orthonormal basis system.

a) What are the components F_{ij} of the deformation gradient $\boldsymbol{F}$?

b) The vectors $\boldsymbol{A} = \boldsymbol{e}_1$, $\boldsymbol{B} = \boldsymbol{e}_2$ and $\boldsymbol{C} = \boldsymbol{e}_3$ describe the edges of a unit cube in the reference configuration. Calculate the coefficients of the deformed vectors $\boldsymbol{a} = \boldsymbol{F}\boldsymbol{A}$, $\boldsymbol{b} = \boldsymbol{F}\boldsymbol{B}$ and $\boldsymbol{c} = \boldsymbol{F}\boldsymbol{C}$ with $\boldsymbol{F}$ from a). Illustrate the deformation for the special case $\alpha = \gamma = 0$ in a sketch.

c) What change in the initial volume V results for $\alpha, \beta, \gamma \neq 0$ and what for the special case $\alpha = \gamma = 0$?

d) Calculate the components of the right Cauchy-Green tensor $\boldsymbol{C} = \boldsymbol{F}^T \boldsymbol{F}$.

e) Compare the Green strain tensor $\boldsymbol{E} = \frac{1}{2}(\boldsymbol{C} - \boldsymbol{1})$ with the classical, geometrically linearized strain tensor $\boldsymbol{\varepsilon}$. Additionally, calculate $\boldsymbol{E}$ and $\boldsymbol{\varepsilon}$ for the deformation gradient from a). What can you determine with regard to the volumetric strain?

Task 2.3: Parameter Determination

Assume that the diagrams shown in Fig. 2.2 were obtained from a uniaxial tensile test. On the left, the tensile stress is plotted against the tensile strain, and on the right, the strains perpendicular to the tensile direction are plotted against the tensile strain.

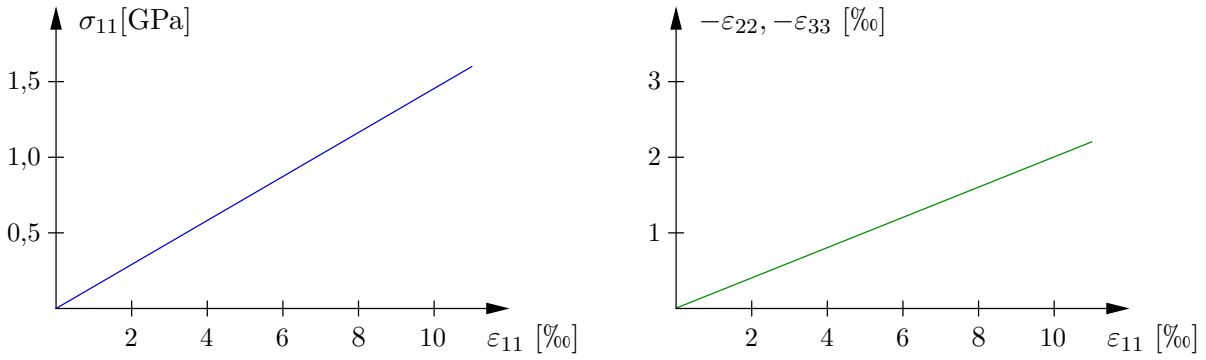

Figure 2.2: Regression lines of the experimental data from a uniaxial tensile test.

Using the constitutive law $\boldsymbol{\sigma} = \lambda\,\mathrm{tr}[\boldsymbol{\varepsilon}]\,\boldsymbol{I} + 2\,\mu\,\boldsymbol{\varepsilon}$ for linear elasticity, determine the Lamé constants λ and μ.